Client Selection under Unreliable Participation: E3CS, Entropic Mirror Descent, and Random Sampling

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Abstract

Unreliable participation reduces the number of updates available in federated learning. Favoring reliable clients can improve participation, but may also concentrate training on a restricted subset of client data. We compare E3CS, an entropic online stochastic mirror-descent (OSMD) selection rule, and uniform random sampling under explicit lower bounds on client inclusion probabilities. Both adaptive methods use binary return feedback and select a fixed number of clients through sampling that preserves their prescribed inclusion probabilities. Experiments on non-IID EMNIST-Letters use 100 clients, 20 selections per round, 400 rounds, and three matched random seeds. Without a fairness floor, OSMD achieves 83.15% mean final accuracy and 6703.3 cumulative successful returns, compared with 82.64% and 6422.7 for E3CS. OSMD also has higher mean final accuracy under the remaining schedules, although the differences are small and not uniformly positive in individual runs. Stronger fixed fairness constraints reduce effective participation without consistently improving final accuracy. These findings identify an empirical advantage for the implemented OSMD rule in the evaluated setting; they do not establish superiority across datasets or changing reliability conditions.

1 Introduction

Federated learning distributes model training across clients while keeping their examples local. In each communication round, a server selects clients, distributes the current model, and aggregates their local updates [1]. This procedure depends on participation as well as statistical quality: a selected client's data cannot influence the model if its update is never returned.

When clients differ in reliability, client selection introduces a practical trade-off. Uniform sampling preserves equal selection opportunities but spends part of the communication budget on unsuccessful participation. Concentrating selections on reliable clients increases the number of usable updates, while potentially restricting the data represented in training. E3CS addresses this trade-off through exponential reward weighting and a minimum per-client selection probability [2].

Our aim is to develop a more effective selection rule for the same problem. We investigate entropic online stochastic mirror descent, which penalizes observed failures and projects its allocation onto the same fairness-constrained feasible set. This provides a direct comparison between additive fairness allocation in E3CS and constrained entropy projection in OSMD. Uniform random sampling supplies a reference that does not learn from participation feedback.

Entropic mirror descent is an established optimization method [3], and OSMD has previously been applied to federated client sampling with a sampling-variance objective [4]. The present work is an implementation and empirical study of a failure-based, fixed-cardinality selection rule, rather than a claim that mirror descent or adaptive client sampling is new.

Our analysis makes the probability model, update rules, sampling procedure, and aggregation convention explicit. The experiments compare methods under shared client data, initialization, local epoch counts, and potential participation outcomes. Reported results are restricted to the completed non-IID EMNIST experiment with FedAvg local training.

2 System model and selection objective

2.1 Unreliable participation

Let $[K] = \{1, \ldots, K\}$ denote the clients and let $A_t \subseteq [K]$ be the selected set in round t, with $|A_t| = k$. The binary variable $x_{i,t}$ indicates whether client i would successfully return its update if selected. The server observes $x_{i,t}$ only for

$i \in A_t$.

For the reported simulation,

$$\begin{aligned} x_{i,t} &\sim \text{Bernoulli}(\rho_i), \\ \rho_i &\in \{0.1, 0.3, 0.6, 0.9\}. \end{aligned} \tag{1}$$

There are 25 clients in each reliability group. Outcomes are generated independently across rounds and clients. Each ρ_i remains fixed throughout training and is unknown to the adaptive selector. Consequently, the experiment models heterogeneous, stationary reliability; it does not evaluate adaptation after reliability changes.

2.2 Inclusion probabilities and fairness

Let $\mathcal{H}_{t-1}$ be the information available before selection and write

$$p_{i,t} = \Pr(i \in A_t \mid \mathcal{H}_{t-1}).$$

These are inclusion probabilities for a set of k distinct clients. They sum to k, rather than one. The feasible allocation set is

$$\begin{aligned} \mathcal{P}_{\sigma_t} = \left\{ p \in \mathbb{R}^K : \sum_i p_i = k,\ \sigma_t \le p_i \le 1 \right\}, \\ 0 \le \sigma_t \le k/K. \end{aligned} \tag{2}$$

The lower bound σ_t guarantees a minimum selection opportunity. At $\sigma_t = 0$ there is no imposed positive floor; at $\sigma_t = k/K$ the feasible set contains only the uniform vector.

This is a constraint on selection, not on successful contribution or prediction quality. Equal inclusion probabilities do not produce equal numbers of returned updates when clients have different ρ_i. Nor does the floor guarantee equal client accuracy or balanced labels in every round.

2.3 Effective participation and model training

The cumulative effective participation is

$$\text{CEP}(T) = \sum_{t=1}^{T} \sum_{i \in A_t} x_{i,t}. \tag{3}$$

Under Eq. (1), a sampler with inclusion vector p_t satisfies

$$\mathbb{E}\left[\sum_{i \in A_t} x_{i,t} \mid \mathcal{H}_{t-1} \right] = \sum_{i=1}^{K} p_{i,t} \rho_i. \tag{4}$$

The adaptive rules use observed returns to favor allocations with higher effective participation, subject to Eq. (2).

The learning objective remains the weighted empirical loss

$$\begin{aligned} f(\theta) &= \sum_{i=1}^{K} \omega_i f_i(\theta), \\ f_i(\theta) &= \frac{1}{|\mathcal{D}_i|} \sum_{(u,v) \in \mathcal{D}_i} \ell(\theta; u, v). \end{aligned} \tag{5}$$

Here $\omega_i = |\mathcal{D}_i| / \sum_j |\mathcal{D}_j|$. Following the reference formulation, effective participation is a surrogate for the effect of selection on learning [2]. It is not equivalent to minimizing Eq. (5): more returned updates can still represent a biased subset of the data.

2.4 Aggregation under unsuccessful participation

Let θ_t be the global model and $B_t = \{i \in A_t : x_{i,t} = 1\}$ the successful selected clients. If $\theta_{i,t}$ is a successful local model, aggregation uses

$$\begin{aligned} \theta_{t+1} &= \sum_{i \in B_t} \omega_i \theta_{i,t} + \sum_{i \notin B_t} \omega_i \theta_t \\ &= \theta_t + \sum_{i \in B_t} \omega_i (\theta_{i,t} - \theta_t). \end{aligned} \tag{6}$$

All clients have 450 training examples, giving $\omega_i = 1/K$. Weights are not renormalized over B_t. Thus, with ten successful clients, each model difference is weighted by 1/100, not 1/10. If B_t is empty, the global model is unchanged. This convention follows the volatile aggregation rule of Huang et al. [2].

3 Selection algorithms

3.1 Uniform random sampling

Random sampling selects a size-k subset uniformly without replacement:

$$\begin{aligned} A_t &\sim \text{Uniform}\{A \subseteq [K] : |A| = k\}, \\ p_{i,t} &= k/K. \end{aligned} \tag{7}$$

It maintains no estimates and does not adapt to return feedback. For $k/K = 0.2$, it satisfies every fairness floor considered here. It is evaluated once per seed and experimental setting because its policy does not depend on the schedule label.

3.2 E3CS

E3CS maintains weights $w_{i,t} > 0$, initialized at $w_{i,1} = 1$. For $\sigma_t < k/K$, its allocation can be

expressed as

$$p_{i,t} = \sigma_t + \min\{1-\sigma_t, c_t w_{i,t}\},$$
$$\sum_i p_{i,t} = k. \tag{8}$$

Here $c_t > 0$ is a normalization scale. Without an active upper bound, this is

$$p_{i,t} = \sigma_t + (k - K\sigma_t)\frac{w_{i,t}}{\sum_j w_{j,t}}.$$

At the maximum quota, the uniform vector is returned directly.

For selected clients, successful return is the reward. The estimate is

$$\widehat{x}_{i,t} = \frac{\mathbf{1}\{i \in A_t\} x_{i,t}}{p_{i,t}}. \tag{9}$$

Define the overflow set $S_t = \{i : c_t w_{i,t} > 1-\sigma_t\}$ and update coefficient $a_t = \eta(k - K\sigma_t)/K$. The weight update is

$$w_{i,t+1} = \begin{cases} w_{i,t}\exp(a_t \widehat{x}_{i,t}), & i \notin S_t, \\ w_{i,t}, & i \in S_t. \end{cases} \tag{10}$$

The overflow-freezing rule follows the reference algorithm [2]. Weights are stored in log form and shifted by their common maximum after updates, preserving their ratios and allocations.

3.3 Entropic online stochastic mirror descent

3.3.1 Failure-based feedback

OSMD treats an unsuccessful return as unit loss:

$$\ell_{i,t} = 1 - x_{i,t},$$
$$\widehat{\ell}_{i,t} = \frac{\mathbf{1}\{i \in A_t\}(1 - x_{i,t})}{p_{i,t}}. \tag{11}$$

A selected failure receives positive estimated loss, while a selected success receives zero. Unselected coordinates also receive zero because their outcomes are not observed; they are not assumed to have succeeded.

If inclusion probabilities are preserved and positive, then $\mathbb{E}[\widehat{x}_{i,t} \mid \mathcal{H}_{t-1}] = \rho_i$ and $\mathbb{E}[\widehat{\ell}_{i,t} \mid \mathcal{H}_{t-1}] = 1 - \rho_i$. The sampling procedure needed for these identities is specified below.

3.3.2 Mirror-descent update

The regularizer is unnormalised negative entropy,

$$F(p) = \sum_i (p_i \log p_i - p_i), \tag{12}$$

with Bregman divergence

$$D_F(p, q) = \sum_i \left[p_i \log \frac{p_i}{q_i} - p_i + q_i \right]. \tag{13}$$

Starting from $p_{i,1} = k/K$, the update solves

$$q_{t+1} = \arg\min_{p \in \mathcal{P}_{\sigma_t}} \left\{ \eta_t \langle p, \widehat{\ell}_t \rangle + D_F(p, p_t) \right\}. \tag{14}$$

Here q_{t+1} denotes the post-update allocation under the current floor. Before the next selection, it is projected onto $\mathcal{P}_{\sigma_{t+1}}$ if the floor changes. This distinction avoids treating a vector feasible for one round as automatically feasible for the next.

Equation (14) has a convenient two-stage form:

$$y_{i,t+1} = p_{i,t}\exp(-\eta_t \widehat{\ell}_{i,t}), \tag{15}$$
$$q_{t+1} = \arg\min_{p \in \mathcal{P}_{\sigma_t}} D_F(p, y_{t+1}). \tag{16}$$

The exponential step reduces the relative allocation of observed failures. The projection restores the fairness floor, upper bounds, and total mass. This is a fixed-cardinality semi-bandit application of entropic mirror descent [3].

3.3.3 Constrained entropy projection

For a positive vector y, consider minimizing $D_F(p, y)$ over $\mathcal{P}_\sigma$. With multipliers λ for the sum constraint and $\mu_i, \nu_i \geq 0$ for the lower and upper bounds, stationarity gives

$$\log(p_i/y_i) + \lambda - \mu_i + \nu_i = 0.$$

Complementary slackness implies that a free coordinate satisfies $p_i = e^{-\lambda} y_i$. Coordinates outside the admissible range are fixed at the corresponding bound. Therefore,

$$p_i = \min\{1, \max\{\sigma, c y_i\}\},$$
$$\sum_i p_i = k. \tag{17}$$

For positive y, strict convexity gives a unique projected vector. The scalar c is found by monotone root search. At $K\sigma = k$, the projection is simply the uniform vector and no root search is needed.

The implementation searches in log space, using 110 bisection steps. It checks the final mass within 10^{-8} and probability bounds within 10^{-10}. Logarithms and divisions use positive floating-point safeguards, and exponential arguments are bounded below by -745. These safeguards address numerical underflow rather than introduce an additional theoretical exploration parameter.

3.3.4 Learning rate

The evaluated OSMD configurations use

$$\eta_t = \eta \frac{k - K\sigma_t}{K}, \qquad \eta = 0.5. \tag{18}$$

This matches the coefficient in the E3CS ex-

ponent within each schedule. It is a controlled experimental choice, not a claim that the two methods have identical sensitivity to their rate. Constant-rate OSMD and a separate base rate are supported but are not evaluated here.

3.4 Why the adaptive rules differ

E3CS allocates a baseline to every client and distributes residual mass through persistent reward weights. OSMD instead updates its current probability vector through estimated losses and projects the result. Only E3CS freezes overflowed weights.

The allocation maps are also distinct. For total mass one, floor 0.1, and weights $(0.6, 0.3, 0.1)$, entropy projection leaves the already feasible vector unchanged, whereas additive allocation gives $(0.52, 0.31, 0.17)$.

Although the true reward and failure objectives differ by the constant k on the feasible set, their estimators satisfy

$$\widehat{\ell}_{i,t} = \frac{\mathbf{1}\{i \in A_t\}}{p_{i,t}} - \widehat{x}_{i,t}.$$

The first term is coordinate dependent and random. It cannot be removed as a common shift in an exponential update. Converting reward to loss therefore does not make the realized E3CS and OSMD updates equivalent.

3.5 Fixed-cardinality sampling

The denominators in Eqs. (9) and (11) must be actual inclusion probabilities. Ordinary weighted sampling without replacement does not generally preserve an arbitrary prescribed vector p_t. The evaluated adaptive configurations therefore use pivotal sampling, rather than the weighted multinomial call used in the reference implementation.

Starting from $q = p_t$, pivotal sampling selects two fractional coordinates q_i, q_j and sets

$$a = \min\{1 - q_i, q_j\}, \qquad b = \min\{q_i, 1 - q_j\}.$$

The pair is replaced by

$$\begin{aligned} (q_i, q_j) &\leftarrow (q_i + a, q_j - a) \quad \text{with prob. } \frac{b}{a+b}, \\ (q_i, q_j) &\leftarrow (q_i - b, q_j + b) \quad \text{with prob. } \frac{a}{a+b}. \end{aligned} \tag{19}$$

Both branches preserve the sum and bounds. Their expected change in q_i is $ab/(a+b) - ba/(a+b) = 0$, and the same holds for q_j. Each operation makes at least one coordinate integral. The integer total mass k then yields exactly k selected clients, while the successive expectation-preserving operations retain the requested marginals.

The routine treats coordinates within 10^{-12} of a boundary as integral and checks the final cardinality. Sampling is dependent within a round, as expected for a fixed-size set; independence is not needed for the first-order importance-weighting identities.

The solver handles E3CS in log-weight space and OSMD in log-probability space. An allocation costs $O(BK)$ with $B = 110$; the implemented pivotal routine costs $O(K^2)$ in the worst case because it scans fractional coordinates repeatedly. Local CNN training dominates the workload.

4 Experimental design

4.1 Data construction and evaluation population

The experiment uses the training partition of EMNIST-Letters with 26 classes. Labels are converted from $1, \ldots, 26$ to $0, \ldots, 25$. For each seed, examples within each class are first separated into global training and test pools, reserving 10% for testing. Client datasets are sampled only after this separation.

Each client receives 450 training examples and 50 test examples. A primary label is selected uniformly for each client. Training sets contain 360 primary-label examples and 90 others; test sets contain 40 primary-label examples and 10 others. Sampling is without replacement within each client split. Different clients may share examples within the training pool or within the test pool, but the training and testing unions are checked to be disjoint.

This prevents a held-out image on one client from entering the global model through another client's training set. It deliberately differs from independently sampled local holdouts. Accuracy is evaluated over the concatenation of all client holdouts, giving 5000 test records. An image shared by several client holdouts is counted at each occurrence. The resulting metric averages over client test records; it is not an evaluation on the official EMNIST test partition or the unique test union.

4.2 Model and local optimization

The network uses two unpadded 5×5 convolution layers with ten output channels each. ReLU and

Table 1: Experimental configuration.

Setting	Value	Setting	Value
Dataset	EMNIST-Letters, non-IID	Local optimizer	SGD; rate 0.01, momentum 0.9, batch size 40
Clients	$K = 100$; $k = 20$ selected per round	Local epochs	Fixed per client; sampled from $\{1, 2, 3, 4\}$
Training horizon	400 communication rounds	Selection rates	Base $\eta = 0.5$; matched OSMD coefficient
Data per client	450 training and 50 test examples	Fairness schedules	0, 0.5, 0.8, and incremental
Reliability groups	25 clients at each of $0.1, 0.3, 0.6, 0.9$	Adaptive sampler	Pivotal; prescribed inclusion marginals
Evaluation	Every round; 5000 client test records	Random seeds	$0, 1, 2$

2×2 max pooling follow each convolution. The spatial dimensions are

$$28 \to 24 \to 12 \to 8 \to 4,$$

giving 160 flattened features. The classifier has widths $160 \to 1280 \to 256 \to 26$, with ReLU after the first two fully connected layers. The final layer returns logits for cross-entropy loss. Pixels are scaled by $1/255$; no augmentation, additional standardization, or orientation correction is applied.

Successful clients train a copy of the same round-start global model. Local SGD uses learning rate 0.01, momentum 0.9, and batch size 40. Momentum is reset for each local training call. Each client's epoch count is sampled once from $\{1, 2, 3, 4\}$ independently of reliability and remains fixed within that seed. Data are reshuffled each epoch; 450 examples give eleven batches of 40 and a final batch of ten.

Potential outcomes are independent of training computation, so the simulator skips local training for failed clients whose updates would be discarded. All successful differences are accumulated before the global update in Eq. (6). This models the same aggregate but does not account for resources consumed by failed work.

4.3 Fairness schedules and evaluation budget

Table 1 summarizes the main settings. Fixed schedule labels $s \in \{0, 0.5, 0.8\}$ correspond to $\sigma_t = sk/K$. The incremental schedule is

$$\sigma_t = \begin{cases} 0, & 1 \le t \le 100, \\ 0.2, & 101 \le t \le 400. \end{cases} \tag{20}$$

After the first quarter, the incremental allocation is necessarily uniform. Earlier differences in training remain in the global model. Later realized selections can also differ because earlier sampling calls may have advanced random generators differently.

The fixed schedules give floors 0, 0.1, and 0.16 and exponent coefficients 0.10, 0.05, and 0.02, respectively. In the incremental schedule, the coefficient is 0.10 for rounds 1–100 and zero afterward. Increasing the floor therefore changes both the feasible set and the update magnitude in this experimental design.

4.4 Matched runs and reproducibility

Each seed specifies a common client assignment, initial model state, local epoch vector, and potential-outcome table for all methods. Local minibatch order is also matched by client and round. This controls several sources of variation without requiring methods to select the same clients or observe the same feedback.

Each seed includes eight adaptive configurations and one random baseline. The complete comparison therefore contains $3 \times 9 = 27$ runs of 400 rounds each. The implementation also supports IID data, CIFAR-10, and FedProx local optimization [5], but no results for these configurations are reported here.

The environment generator uses seed $s+987654$ and each selector's sampling generator begins from $s + 123456$. The local shuffle seed for zero-based round r and client i is

$$(s \cdot 1000003 + rK + i) \bmod (2^{63} - 1).$$

This matches minibatch order for the same client and round without forcing different methods to follow identical training trajectories.

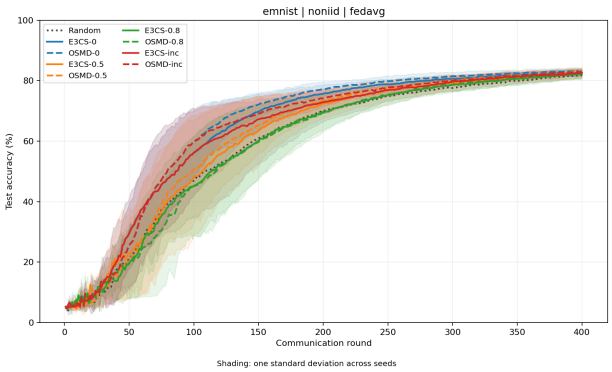


Figure 1: Test accuracy on non-IID EMNIST-Letters with FedAvg. Lines are means over three seeds; bands show one sample standard deviation. Solid lines represent E3CS, dashed lines OSMD, and the dotted gray line random sampling. Colors distinguish fairness schedules.

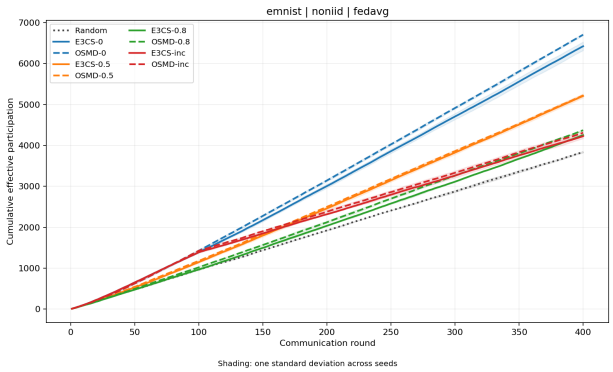


Figure 2: Cumulative effective participation for the same configurations. Means and shaded sample standard deviations are calculated across three seeds. CEP counts successful selected clients; it does not directly measure the quality or representativeness of their updates.

5 Results

5.1 Evaluation measures

We report final test accuracy, CEP, and cumulative success ratio $\mathrm{SR}(T) = \mathrm{CEP}(T)/(Tk)$. Target-round measurements are the first evaluated rounds reaching 65%, 75%, or 85% accuracy. An unreached target is left missing, rather than replaced by the final round. Every-round evaluation makes these exact first crossings within the recorded run.

Aggregate results use the arithmetic mean and sample standard deviation across the three seeds. Shaded bands in the figures have the same interpretation; they are not confidence intervals. All tables are calculated from the saved seed-level measurements.

5.2 Learning and participation trajectories

Figure 1 shows substantial learning over 400 rounds, with appreciable variation across seeds during intermediate training. OSMD-0 generally follows the strongest mean accuracy trajectory. The incremental policies initially coincide with their respective no-floor policies, then continue from those learned models under uniform inclusion probabilities.

The participation curves in Figure 2 separate the policies more clearly. Removing the floor permits the largest gains in successful returns. The 0.5 setting follows, while the 0.8 and incremental settings leave less room for sustained preference toward reliable clients. The incremental curves change slope after round 100, consistent with Eq. (20).

5.3 Final performance

Table 2 reports final accuracy and participation. OSMD-0 achieves the largest mean final accuracy, 83.147%, and CEP, 6703.3. Compared with E3CS-0, this is 0.507 percentage points of accuracy and 280.7 additional successful returns, corresponding to 4.37% more CEP. Relative to random sampling, the gains are 1.367 percentage points and 2867.0 returns.

Both adaptive methods lose CEP as the fixed floor increases. This is consistent with reserving more allocation for clients with lower reliability. The strongest fixed floor does not compensate through higher mean accuracy: E3CS-0.8 and OSMD-0.8 finish below their no-floor counterparts. The incremental schedule improves on the respective 0.8 configuration in mean accuracy but does not surpass OSMD-0.

As a consistency check, uniform sampling has expected returns per round

$$\frac{k}{K}\sum_i \rho_i = 0.2 \times 25(0.1 + 0.3 + 0.6 + 0.9)$$
$$= 9.5.$$

Its expected CEP is 3800, close to the observed mean of 3836.3. This check concerns the participation simulation, rather than a claim about the expected learning accuracy.

5.4 Time to accuracy targets

All runs reach 65% and 75% accuracy; none reaches 85% within the budget. The target-round

Table 2: Mean final accuracy and CEP with sample standard deviations, and mean first-crossing round at 75%. Higher accuracy and CEP are better; fewer target rounds are better.

Method	Schedule	Accuracy (%)	CEP	Round@75
Random	uniform	81.780 ± 1.054	3836.3 ± 17.1	249.3
E3CS	0	82.640 ± 1.115	6422.7 ± 104.0	185.0
OSMD	0	83.147 ± 1.093	6703.3 ± 19.6	174.3
E3CS	0.5	82.440 ± 1.561	5208.0 ± 47.8	228.7
OSMD	0.5	82.973 ± 1.436	5221.7 ± 33.5	211.3
E3CS	0.8	82.120 ± 1.593	4251.3 ± 19.3	245.0
OSMD	0.8	82.287 ± 1.171	4370.3 ± 8.4	241.0
E3CS	inc	82.527 ± 0.991	4226.3 ± 54.6	220.7
OSMD	inc	82.947 ± 1.035	4310.7 ± 41.0	201.3

column in Table 2 averages individual first crossings, which need not equal the crossing of an average curve.

OSMD-0 reaches 75% in 174.3 rounds on average, compared with 185.0 for E3CS-0 and 249.3 for random sampling. OSMD reaches this target earlier on average under each matched schedule. The same ordering does not hold at every target: OSMD-0.8 reaches 65% about 1.7 rounds later than E3CS-0.8, while reaching 75% four rounds earlier.

5.5 Paired differences

To compare the adaptive methods under shared conditions, we subtract E3CS from OSMD within each matching seed and schedule. The mean final-accuracy differences are +0.507, +0.533, +0.167, and +0.420 percentage points for schedules 0, 0.5, 0.8, and inc, respectively. All favor OSMD, but their magnitudes are modest.

The seed-level outcomes qualify the average ordering. At schedule 0.8, OSMD's accuracy differences are +0.24, −0.40, and +0.66 percentage points. Its CEP differences at schedule 0.5 are −33, +13, and +61. Thus neither every metric nor every matched run favors OSMD. With three seeds, we treat the differences as descriptive evidence, without a statistical-significance claim. The marginal standard deviations in Table 2 are not standard deviations of paired differences.

6 Discussion

Participation feedback is useful in the evaluated client population: every adaptive configuration increases mean CEP over random sampling. OSMD-0 combines the largest participation gain with the strongest mean accuracy and target-round results. However, differences in final accuracy are much smaller than differences in CEP. Successful return counts measure the availability of updates, while learning also depends on the client data, local optimization, and the direction of those updates.

The fairness results are similarly conditional. A larger floor restricts allocation toward reliable clients, but does not consistently improve final accuracy here. Under the matched-rate design it also reduces the exponent coefficient. A separate sensitivity study would be needed to distinguish changes in feasible allocation from changes in update rate.

Several limits constrain interpretation. First, the completed experiments cover one dataset, one non-IID construction, and one local training scheme. Second, only three seeds are available, and each varies client assignment, initialization, local epochs, and participation outcomes jointly. Third, stationary reliability does not test recovery after failures or availability patterns change. Fourth, there is no explicit communication- time or energy model, so fewer rounds cannot be converted directly into wall-clock or energy savings.

Global holdout separation removes cross-client train/test leakage, but evaluation still describes the sampled client population and includes repeated held-out images across clients. Finally, no regret bound or federated convergence guarantee is derived for the combined procedure with multiple selections, probability caps, and changing quotas. The observed curves do not substitute for such an analysis.

7 Conclusion

We compared E3CS, a failure-based entropic OSMD selection rule, and uniform random sampling under unreliable participation. The comparison uses fixed-cardinality sampling with controlled inclusion probabilities and the reference

aggregation convention. It separates the reward-weighted, additive allocation of E3CS from OSMD's loss update and constrained entropy projection.

On non-IID EMNIST with FedAvg and three matched seeds, OSMD has higher mean final accuracy than the corresponding E3CS configurations under all reported schedules. Its largest participation advantage occurs without a fairness floor. The accuracy gains remain modest, with exceptions at individual seeds and target levels. These results favor the implemented OSMD rule in this setting, while leaving generalization to other data distributions, rate choices, and changing client reliability unresolved.

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